

Week 3 · Day 3 · Route Tables, Internet Gateways & NAT

Configure Security Groups — Your VPC Firewall

 Time

60 min

 Objective

Create Security Groups for a 3-tier web application (Load Balancer, Web Servers, Database) with correct inbound/outbound rules using security group chaining.

Prerequisites — Complete Before Starting

Custom VPC from Week 3 Days 1-2

Portfolio Goal — What You Will Post

Add a Security Group rules table to your portfolio — shows you can design stateful firewalls for real applications.

Lab Steps

Step 1 Understand Security Groups vs NACLs

1. Before building: write down the 5 key differences between Security Groups and NACLs.
2. Security Groups: stateful, instance-level, allow rules only, evaluated as a group.
3. NACLs: stateless, subnet-level, allow AND deny rules, evaluated in order.
4. In most architectures, Security Groups do the heavy lifting. NACLs are an extra layer.

Step 2 Create the Load Balancer Security Group

1. Go to EC2 → Security Groups → Create security group.
2. Name: SG-LoadBalancer | Description: Allow HTTPS from internet.
3. VPC: OluTech-Production-VPC.
4. Inbound rules: Type: HTTPS | Protocol: TCP | Port: 443 | Source: 0.0.0.0/0.
5. Inbound rules: Type: HTTP | Protocol: TCP | Port: 80 | Source: 0.0.0.0/0.
6. Outbound rules: Keep default (All traffic allowed outbound).
7. Click Create security group.

Step 3 Create the Web Server Security Group

1. Create security group: SG-WebServers | Description: Allow traffic from Load Balancer only.
2. VPC: OluTech-Production-VPC.
3. Inbound rules: Type: HTTP | Port: 80 | Source: CUSTOM → select SG-LoadBalancer.
4. This is security group chaining — traffic is only allowed from the load balancer SG.
5. Inbound rules: Type: SSH | Port: 22 | Source: YOUR IP ONLY (get it from whatismyip.com — format: x.x.x.x/32).
6. Click Create security group.

Step 4 Create the Database Security Group

1. Create security group: SG-Database | Description: Allow MySQL from Web Servers only.
2. VPC: OluTech-Production-VPC.
3. Inbound rules: Type: MySQL/Aurora | Port: 3306 | Source: CUSTOM → select SG-WebServers.
4. NO internet access, NO SSH. Only the web servers can talk to the database.
5. Click Create security group.

Step 5 Document Your Security Group Architecture

1. Create a table in your notes: SG Name | Inbound Rules | Outbound Rules | Connected To.
2. Draw a diagram showing the chaining: Internet → SG-LoadBalancer → SG-WebServers → SG-Database.
3. Write next to each arrow: what port/protocol is allowed.
4. Export diagram. Save to portfolio.

Screenshot Checklist

Take these screenshots and save them to your portfolio folder:

- ☐ SG-LoadBalancer with HTTP/HTTPS inbound rules
- ☐ SG-WebServers with port 80 from SG-LoadBalancer (chained)
- ☐ SG-Database with port 3306 from SG-WebServers (chained)
- ☐ Your security group architecture diagram

Portfolio Post — Copy & Customise

LinkedIn: 'Security group chaining — the professional way to configure AWS firewalls. Instead of allowing 0.0.0.0/0 everywhere, each tier only accepts traffic from the previous tier (using SG-to-SG rules). Zero internet exposure for the database. Architecture diagram below. #AWSsecurity #VPC #CloudNetworking'

Bonus Challenge — Push Your Skills Further

Create a Network ACL for your private subnets. Add an explicit DENY rule for inbound traffic on port 23 (Telnet) and port 21 (FTP) — two insecure protocols. Add an ALLOW rule for ports 3306 (MySQL). Test how the NACL evaluation order matters.

Lab Complete — Mark Off Your Progress

Lab Steps Completed	☐
All Screenshots Taken	☐
Portfolio Artifact Posted	☐
Bonus Challenge Attempted	☐